Coláiste na Tríonóide, Baile Átha Cliath
Trinity College Dublin
Ollscoil Átha Cliath | The University of Dublin

Faculty of Engineering, Mathematics and Science
School of Mathematics

SS Mathematics **Trinity Term 2021**
SS Theoretical Physics

MAU44400 Quantum Field Theory

Wednesday, 12 May **12pm - 6pm (6 hours)**

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Instructions that apply to all take-home exams:

1. This is an open-book exam. You are allowed to use your class notes, textbooks and any material which is directly linked from the module's Blackboard page or from the module's webpage, if it has one. You may not use any other resources except where your examiner has specifically indicated in the "Additional instructions" section below. Similarly, you may only use software if its use is specifically permitted in that section. You are not allowed to collaborate, seek help from others, or provide help to others.

2. If you have any questions about the content of this exam, you may seek clarification from the lecturer using the e-mail address provided. You are not allowed to discuss this exam with others. You are not allowed to send exam questions or parts of exam questions to anyone or post them anywhere.

3. Unless otherwise indicated by the lecturer in the "Additional instructions" section, solutions must be submitted through Blackboard in the appropriate section of the module webpage by the deadline listed above. You must submit a single pdf file for each exam separately and sign the following declaration in each case. It is your responsibility to check that your submission has uploaded correctly in the correct section.

Additional instructions for this particular exam:

Credit will be given for the best two questions from Part A (Questions 1, 2 and 3) and the best two questions from Part B (Questions 4, 5 and 6).

Plagiarism declaration: I have read and I understand the plagiarism provisions in the General Regulations of the University Calendar which are available through https://www.tcd.ie/calendar.

Signature: _____

PART A

1. Noether current and massive electromagnetic field: 25 Marks

According to the Noether theorem, a translationally invariant system of classical fields ϕ_a has a conserved stress-energy tensor:

$$T^{\mu\nu}_{\text{Noeth}} = \sum_a \frac{\partial \mathcal{L}}{(\partial_\mu \phi_a)} \partial^\nu \phi^a - \eta^{\mu\nu} \mathcal{L}$$

To assure that stress-energy tensor is symmetric in μ, ν (which is necessary for the angular momentum conservation) one sometimes has to add a total derivative:

$$T^{\mu\nu} = T^{\mu\nu}_{\text{Noeth}} + \partial_\lambda \mathcal{K}^{[\lambda\mu]\nu}$$

where $\mathcal{K}^{[\lambda\mu]\nu}$ is some 3–index Lorentz tensor antisymmetric in its first two indices.

a) 5 Marks: Show that regardless of the specific form of $\mathcal{K}^{[\lambda\mu]\nu}(\phi, \partial\phi)$

$$\partial_\mu T^{\mu\nu} = \partial_\mu T^{\mu\nu}_{\text{Noeth}} = 0$$

$$P^\mu \equiv \int d^3x T^{0\mu} = \int d^3x T^{0\mu}_{\text{Noeth}}$$

b) 10 Marks: Consider the free electromagnetic field described by the Lagrangian

$$\mathcal{L}(A_\mu, \partial_\nu A_\mu) = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}; \qquad F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu$$

Write down $T^{\mu\nu}_{\text{Noeth}}$ for this theory and show that it is neither symmetric nor gauge invariant.

c) 10 Marks. Properly symmetric, and also gauge invariant, stress-energy tensor is:

$$T^{\mu\nu}_{\text{EM}} = -F^{\mu\lambda} F^{\nu}_{\ \lambda} + \frac{1}{4} \eta^{\mu\nu} F_{\kappa\lambda} F^{\kappa\lambda}$$

Show that this expression indeed has above form for some $\mathcal{K}^{[\lambda\mu]\nu}$ and find $\mathcal{K}^{[\lambda\mu]\nu}$.

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2. $SO(N)$ invariant scalar field theory: 25 marks

Consider an $SO(N)$ symmetric theory of N real scalar fields with Lagrangian:

$$\mathcal{L} = \frac{1}{2}\partial_\mu \Phi^a \partial^\mu \Phi^a - \frac{1}{2}m^2 \Phi^a \Phi^a - \frac{1}{4}\lambda(\Phi^a \Phi^a)^2$$

a) 5 marks: Derive classical field equations

b) 5 marks: Introduce conjugate momenta $\Pi^a(x)$ with Poisson brackets

$$\{\Pi^a(x), \Phi^b(y)\} = \delta^{ab}\delta^{(3)}(x-y)$$

and derive the Hamiltonian density $\mathcal{H}(\Pi, \Phi)$ as a functional of Π^a and Φ^a.

c) 5 marks: Write down the Noether currents J^μ_{ab} of the $SO(N)$ symmetry: $\Phi^a \to \Lambda^{ab}\Phi^b$, or infinitesimally $\Phi^a \to \Phi^a + \epsilon^{ab}\Phi^b$, with $\Lambda^T\Lambda = I, \quad \Lambda = I + \epsilon, \quad \epsilon^{ab} = -\epsilon^{ba}$.

d) 5 marks: Using classical equations of motion, derived in a), show that these currents are conserved: $\partial_\mu J^\mu_{ab} = 0$

e) 5 marks: Write $SO(N)$ Noether currents J^μ_{ab} in terms of Π^a and Φ^a; check that Noether charges $Q_{ab} = \int d^3x J^0_{ab}(x)$ generate above $SO(N)$ transformations on fields under Poisson Bracket - this means that you need to show that:

$$\delta\Phi^a = \{\sum_{b,c} \epsilon^{bc} Q_{bc}, \Phi^a(x)\} = \epsilon^{ab}\Phi^b$$

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3. Holomorphic quantization: 25 marks

Consider bosonic commutation relations:

$$[a_\alpha, a_\beta] = [a^\dagger_\alpha, a^\dagger_\beta] = 0$$

$$[a_\alpha, a^\dagger_\beta] = \delta_{\alpha\beta}$$

a) 5 marks: Calculate commutators:

$$[a^\dagger_\alpha a_\beta, a^\dagger_\gamma]; \qquad [a^\dagger_\alpha a_\beta, a_\delta]; \qquad [a^\dagger_\alpha a_\beta, a^\dagger_\gamma a_\delta]; \qquad [a^\dagger_\alpha a^\dagger_\beta a^\dagger_\gamma a_\delta, a^\dagger_\mu a_\nu]$$

b) 5 marks: For single pair of a and $a^\dagger$ show that for any holomorphic function

$$f(x) = f_0 + f_1 x + f_2 x^2 + ... :$$

$$[a, f(a^\dagger)] = \frac{\partial}{\partial a^\dagger} f(a^\dagger) = f'(a^\dagger)$$

$$[a^\dagger, f(a)] = -\frac{\partial}{\partial a} f(a) = -f'(a)$$

c) 5 marks: Show that for any constant number c:

$$e^{ca} a^\dagger e^{-ca} = a^\dagger + c$$

$$e^{ca^\dagger} a e^{-ca^\dagger} = a - c$$

$$e^{ca} f(a^\dagger) e^{-ca} = f(a^\dagger + c)$$

$$e^{ca^\dagger} f(a) e^{-ca^\dagger} = f(a - c)$$

d) 10 marks: Now generalise for any number of creation and annihilation operators:

$$\exp(\sum_\alpha c_\alpha a_\alpha) f(a^\dagger) \exp(-\sum_\alpha c_\alpha a_\alpha) = f(a^\dagger_\alpha + c_\alpha)$$

$$\exp(\sum_\alpha c_\alpha a^\dagger_\alpha) f(a) \exp(-\sum_\alpha c_\alpha a^\dagger_\alpha) = f(a_\alpha - c_\alpha)$$

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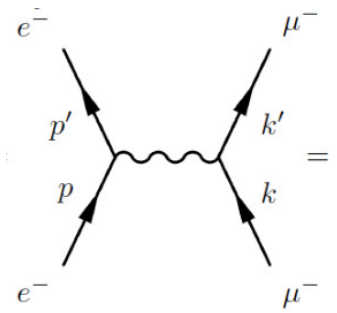
PART B

4. $e^-\mu^- \to e^-\mu^-$ **tree level: 25 marks**

a) 15 marks: Calculate spin-averaged amplitude squared of the scattering of an electron with a muon ($\frac{1}{4}\sum|\mathcal{M}|^2$):



$$= i\mathcal{M}$$

b) 10 marks: Take the limit where muon mass m_μ is large and simplify the formula (limit when $\frac{m_e}{m_\mu} \to 0$).

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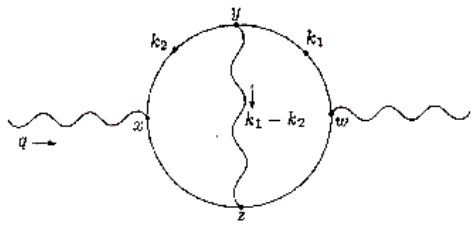
5. QED two-loops: 25 Marks

a) 15 marks: Write the momentum space integral representation for following two-loop diagram in QED:



b) 10 marks: Rewrite the expression in terms of Feynman parameters. Do not need to calculate the integrals or traces, only representing in terms of the integration over Feynman parameters and undetermined momenta is required.

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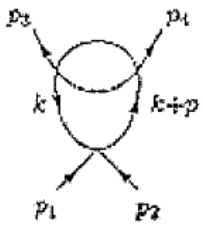
6. 2-loop in ϕ^4 theory: 25 Marks

a) 10 Marks: Write down the expression for the following 2-loop Feynman diagram $V_{[2-loop]}$ in $\frac{1}{4}\lambda\phi^4$ theory



in terms of the integral over undetermined loop momentum k and 1-loop 4-point sub-diagram $V_{[1-loop]}$ (with outgoing momenta $[p_3, p_4]$ and incoming momenta $[-k, k+p]$).

b) 5 Marks: Express $V_{[1-loop]}$ in terms of the integral over its own loop momentum.

c) 5 Marks: Use Feynman parameter representation, separately for $V_{[1-loop]}$ and $V_{[2-loop]}$ (which has been already written in a) in terms of $V_{[1-loop]}$); do not have to calculate the integrals.

d) 5 Marks: Is this 2-loop diagram divergent or convergent? Explain the answer.

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